Part II, Lecture 4: Loan Pricing and Other Applications of Credit Risk Measures

By the end of this lecture you must know why ratings are a competitive weapon against adverse selection, how a price-setting bank derives the minimum loan rate from the risk-adjusted equilibrium condition, how the same condition rearranged gives RAROC and the accept/reject rule against the cost of equity, how EVA expresses value creation in euros, how ratings set credit approval authority through expected loss and VaR limits, and how ratings drive regulatory capital, lending volume and pro-cyclicality.

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1. Ratings as a Competitive Tool

Lending is the bank’s core business, and credit risk is its key challenge: who to lend to, how much, at what rate. The answer runs through credit risk measurement, primarily ratings. The same rating plays two different roles depending on who sets the price.

- **Price setter:** the bank uses ratings to differentiate pricing. Better ratings get lower rates, worse ratings get higher rates. Accurate pricing attracts better clients and leaves the riskier, less profitable deals to less sophisticated competitors.
- **Price taker:** the bank must accept market rates. Ratings then evaluate whether the market rate adequately covers the risk: reject the loan when the risk is high relative to the rate, accept when it is low.

2. The Adverse Selection Risk

A bank without robust rating systems lacks accurate risk information and prices on average risk or on cruder methods. Two flows follow: good borrowers leave for banks that reward their low risk with better rates, and bad borrowers arrive, attracted by underpriced high risk. The unsophisticated bank ends up with a disproportionate share of bad loans and significant losses.

Ratings are a defence against adverse selection: without them, the bank’s own pricing systematically selects the worst borrowers against it.

3. Loan Pricing: the Risk-Adjusted Equilibrium

Take a 1-year loan of 1 euro at rate iL. The gross repayment due is 1 + iL; net of expected loss, the expected repayment is (1 + iL) x (1 - PD x LGD). For the loan to make sense, this expected repayment must cover the principal, the funding cost of the debt portion, the required return on the equity portion, and operational costs.

Equilibrium condition:
(1 + iL) x (1 - PD x LGD) = 1 + IFTR x (1 - VaR) + Ke x VaR + C
where IFTR is the internal fund transfer rate (cost of debt funding, set by treasury), VaR the capital allocation for the loan’s risk (so 1 - VaR is the debt-funded portion), Ke the cost of equity required by shareholders, and C operational costs as a percentage of the loan.

Solving for the rate gives the minimum interest rate covering all costs and the target return on equity, adjusted for credit risk:

Minimum loan rate: iL = [PD x LGD + IFTR x (1 - VaR) + Ke x VaR + C] / (1 - PD x LGD)

Read the numerator as a sum of risk-adjusted cost components: expected loss, debt funding on the debt portion, equity remuneration on the capital portion, and operations. The denominator lifts the rate because a share PD x LGD of the

promised repayment is expected never to arrive, so the paying borrowers must compensate.

4. Worked Example: the Minimum Loan Rate

The slides' example: PD = 10% (borrower rating), LGD = 40% (severity rating), IFTR = 7% (treasury), VaR = 9% (credit risk models), Ke = 14% (shareholders), C = 1% (cost control).

PD x LGD = 0.10 x 0.40 = 0.04
IFTR x (1 - VaR) = 0.07 x 0.91 = 0.0637
Ke x VaR = 0.14 x 0.09 = 0.0126
C = 0.01
iL = (0.04 + 0.0637 + 0.0126 + 0.01) / (1 - 0.04) = 0.1263 / 0.96 = 13.16%

Each input has an owner inside the bank: the borrower rating supplies PD, the severity rating LGD, the CFO and treasury the fund transfer rate, the credit risk models the VaR, the CEO and shareholders the cost of equity, and cost control the operational cost. Pricing is where all these functions meet.

5. How Ratings Enter the Pricing Formula

- **Directly:** PD comes from the borrower rating, LGD from the severity rating.
- **Indirectly:** the capital allocation, whether measured as VaR (economic capital) or as MRC (minimum required capital under the Standardized approach or the Basel risk-weighting functions, which themselves take PDs and LGDs as inputs).

So one borrower's rating moves several inputs of iL at once. Can two banks charge different rates for the same borrower, the same loan, at the same time? Yes: different funding costs, different capital models, different target returns and different cost structures all legitimately produce different equilibrium rates, which is precisely why the more sophisticated bank wins the better borrowers.

6. RAROC: the Price Taker's View

Two situations call for the same formula read backwards: the market sets the rate and the bank must decide whether to grant the loan, or the loan is already granted and the bank wants to measure achieved results. Solving the equilibrium condition for Ke instead of iL yields RAROC, the risk-adjusted return on capital (also named ROCA, RORAC or RARORAC depending on the tradition).

RAROC = [iL - IFTR x (1 - VaR) - PD x LGD x (1 + iL) - C] / VaR

Numerator: the return generated by the loan minus the cost of debt funding, the expected loss on principal and interest, and operational costs, that is the profit available to compensate equity holders. Denominator: the economic capital absorbed by the transaction, the measure of its unexpected loss.

7. Decision Rule and Economic Value Added

Decision rule: compare RAROC with the minimum required return on equity Ke.

- **RAROC > Ke:** the loan earns more than shareholders require: value creation, accept if other factors permit.
- **RAROC < Ke:** value destruction in economic terms, even if the loan is profitable in accounting terms: reject.

EVA in euros: EVA = (RAROC - Ke) x VaR. Positive when RAROC exceeds Ke (value created), negative below (value destroyed), zero at exactly Ke (value maintained). EVA translates the rate spread into an absolute monetary amount.

Check the coherence with the worked example: a loan priced at exactly iL = 13.16% delivers a numerator of 0.0126 on a capital of 0.09, that is RAROC = 14% = Ke and EVA = 0, as it must be for a rate built to just cover the cost of equity.

8. RAPMs for Strategic Decisions

Used ex post, RAROC and EVA are risk-adjusted performance measures. They evaluate the risk-adjusted profitability of geographic areas, product lines (mortgages, cards), customer segments (SME, corporate, retail) and banking divisions (corporate and investment banking, private banking, consumer credit, leasing and factoring). Strategically, RAPMs drive capital allocation toward the most value-creating activities, strategic planning about where to grow and where to exit, and ultimately the maximisation of overall bank value, with a consistent risk-adjusted view across diverse activities.

9. Credit Approval Authority

How large a loan may a given officer approve? The traditional answer, exposure size in euros, is simple but ignores risk: a 1M loan to a high-risk borrower can be riskier than 10M to a low-risk one. The modern approach delegates on risk metrics derived from ratings, aligning authority with the risk actually taken.

- **Delegation by expected loss limit EL*:** each rank gets a maximum EL per transaction or customer. Since EL = PD x LGD x EAD, the maximum approvable exposure for a given borrower is EAD = EL* / (PD x LGD). Higher PD or LGD shrinks the maximum exposure under the same limit.
- **Delegation by unexpected loss limit VaR'*:** authority linked to a maximum gross value at risk, EAD = VaR'* / LRCL, where LRCL is the loss rate at the chosen confidence level from credit risk models, incorporating PD, LGD, volatility and possibly correlation. Higher risk again means lower maximum exposure.
- **From EAD back to the current exposure:** since EAD = Ec x Em, the maximum current exposure an officer can grant is Ec = EAD / Em. A committed line with low current usage can hide a high EAD through its multiplier.

10. Ratings, Regulatory Capital and Pro-cyclicality

Ratings also bound the total volume of lending. Under Basel regulation, minimum required capital must satisfy MRC >= solvency coefficient x total RWA (for instance 8%). Under the Internal Ratings-Based approach, a loan's risk-weighted assets are computed from its risk parameters, PD, LGD and maturity, which the internal ratings drive: higher risk means a higher risk weight and higher RWA for the same loan.

The chain: worse ratings, higher RWA, higher minimum required capital, and with capital fixed in the short term, the bank must cut assets to preserve its capital ratio, which in practice means cutting loans. This chain is **pro-cyclical**: in downturns PDs and LGDs rise, RWA rises, and banks are forced to restrict credit precisely when the economy most needs it, deepening the slump. Basel III mitigates the effect, in particular through the **countercyclical capital buffer**.

11. Customer Relationships: the Limits of Automation

Ratings improve customer relationship management and let the bank grow the loan portfolio consistently with its loan policy. The slides close with Mario Draghi's caution: statistical scoring loses part of its predictive power in exceptional times, and credit decisions should integrate model outputs with direct knowledge of the customer, of growth and profitability potential in the long term. Local roots are valuable and should be rebuilt where lost; on-the-ground knowledge should be maximised, avoiding an excess of automation.

12. Common Mistakes and Exam Traps

- Mixing up the two roles of the same rating: price setter (differentiate rates by rating) versus price taker (test the market rate against the risk). The exam often describes the bank's situation in one line; read who sets the price before choosing the formula's direction.
- Forgetting the denominator of the pricing formula: iL is the numerator of costs divided by (1 - PD x LGD), because part of the promised repayment never arrives. Writing iL = PD x LGD + IFTR x (1 - VaR) + Ke x VaR + C without dividing understates the rate.
- Computing RAROC with the wrong expected-loss term: in the RAROC numerator the expected loss is PD x LGD x (1 + iL) (loss on principal AND interest), not PD x LGD alone.
- Rejecting a loan that is profitable in accounting terms when RAROC < Ke: the rule is economic, a loan below Ke destroys value even with positive net income.
- Treating EVA = 0 as a bad outcome: at RAROC = Ke the loan exactly covers the cost of equity, value is maintained; only EVA < 0 destroys value.
- Approving on exposure size alone: the modern delegation converts risk limits into exposure through EAD = EL* / (PD x LGD) or EAD = VaR'* / LRCL, then Ec = EAD / Em; the same officer can approve a much larger amount for a better-rated borrower.
- Forgetting the committed-line multiplier: a line with low current usage can carry a high EAD through Em, so current exposure understates the real risk taken.
- Reversing the pro-cyclical chain: it runs ratings worse, RWA up, MRC up, lending cut; and the Basel III mitigant is the countercyclical capital buffer, not a lower solvency coefficient.
- Believing two banks must quote the same rate for the same borrower: different funding costs, capital models, target returns and cost structures legitimately produce different equilibrium rates.

13. Must Know Absolutely

Must know:
• Price setters differentiate rates by rating; price takers test the market rate against the risk. Ratings defend against adverse selection.
• Equilibrium: (1 + iL)(1 - PD x LGD) = 1 + IFTR x (1 - VaR) + Ke x VaR + C, so iL = [PD x LGD + IFTR(1 - VaR) + Ke x VaR + C] / (1 - PD x LGD).
• Slides' numbers (PD 10%, LGD 40%, IFTR 7%, VaR 9%, Ke 14%, C 1%) give iL = 13.16%.
• Ratings enter directly through PD and LGD, indirectly through VaR or MRC.
• RAROC = [iL - IFTR(1 - VaR) - PD x LGD(1 + iL) - C] / VaR; accept when RAROC > Ke, reject below; EVA = (RAROC - Ke) x VaR.
• Ex post, RAROC and EVA are RAPMs for areas, products, segments and divisions: capital allocation and strategic

planning.
• Approval limits on risk: EAD = EL/*(PD x LGD) or EAD = VaR'*/LRCL, then Ec = EAD/Em.
• Regulatory chain: ratings, RWA, MRC, lending volume, pro-cyclical; Basel III answers with the countercyclical capital buffer.